

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 8: Introduction to Trigonometry

Previous year board questions, topic-wise: 11 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Ratios of standard angles

1. Find the acute angle θ for which $7 + 4 \sin \theta = 9$. **CBSE 2025, 1 mark**
2. Given that $x \cos 60^\circ + y \cos 0^\circ + \sin 30^\circ - \cot 45^\circ = 5$, find the value of $x + 2y$. **CBSE 2025, 2 marks**
3. Evaluate $\frac{\tan^2 60^\circ}{\sin^2 60^\circ + \cos^2 30^\circ}$. **CBSE 2025, 2 marks**
4. Given $\cos(\alpha + \beta) = 0$, find $\cos\left(\frac{\alpha + \beta}{2}\right)$. (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{2}$ (c) 0 (d) $\sqrt{2}$ **CBSE 2024, 1 mark**
5. Find the value of $2\sqrt{2} \cos 45^\circ \sin 30^\circ + 2\sqrt{3} \cos 30^\circ$. **CBSE 2024, 2 marks**
6. For $A = 60^\circ$ and $B = 30^\circ$, verify that $\sin(A + B) = \sin A \cos B + \cos A \sin B$. **CBSE 2024, 2 marks**

2 Trigonometric identities

7. Simplify $\tan^2 \theta - \left(\frac{1}{\cos \theta} \times \sec \theta\right)$. **CBSE 2025, 1 mark**
8. Prove that $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2025, 3 marks**
9. Prove that $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2 \sin^2 A - 1}$. **CBSE 2025, 3 marks**
10. Given $\sec \theta - \tan \theta = m$, find $\sec \theta + \tan \theta$. (a) $1 - \frac{1}{m}$ (b) $m^2 - 1$ (c) $\frac{1}{m}$ (d) $-m$ **CBSE 2024, 1 mark**
11. Prove the identity $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$. **CBSE 2024, 3 marks**

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